Theo Glashier

PhD candidate at Imperial College aspiring to advance research in infrastructure health monitoring

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Imperial College profile

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Profile

I am an early career researcher focused on promoting structural health monitoring for society's ageing critical infrastructure. I work on data-driven strategies that prepare structural datasets for analysis and overcome the pervasive effect of environmental and operational variability.

I wish to pursue a career in research in the long-term monitoring of structures to enable real-time condition assessments to optimise maintenance budgets and workflows for civil infrastructure.

Education

PhD Candidate, Imperial College London, Civil and Environmental Engineering
March 2021 - December 2024

- Thesis entitled 'Temperature-based measurement interpretation of ageing critical infrastructure' and supervised by Dr Craig Buchanan
- Developed specialty expertise in data-driven SHM techniques, machine learning, high performance computing and programming in Python and Matlab
- Completed multiple work trips to meet collaborators, for conferences as well as to undertake operational testing on the MX3D Bridge in Amsterdam
- Gained excellent written and oral communication skills through conferences, poster and presentation competitions, presenting to industry and in seminars, as well as numerous teaching experiences
- Have co-supervised 8 MEng and MSc students throughout my PhD Studies, gaining first-hand experience in how to best guide students' learning and leading them to publishing their work

MEng Mechanical Engineering, University of Sheffield, First-Class Honours
October 2015 - July 2019

- Thesis entitled 'Operational modal analysis for nonlinear systems,' and supervised by Prof. Keith Worden (grade: 1^{st})
- Undertook a third year project (equivalent to a BEng dissertation) in the design and specification of a wind turbine bearing test rig that lead to a research summer internship with RGL Associates
- Developed specialty expertise in structural dynamics, condition monitoring, material integrity, solid mechanics (grades: 82%, 89%, 80%, 75%, respectively) and general mechanical engineering skills

Non-academic appointments

E-monitoring engineer graduate scheme, Total Energies, Pau, France
October 2019 - September 2020

- Developed excellent project management skills through delivering monitoring projects for offshore energy infrastructure within a global enterprise
- Gained invaluable experience in delivering professional technical presentations to internal clients
- Took the technical lead on a CO_2 emission monitoring project, building the data infrastructure template in PI System and applying it to a large portion of Total Energies' worldwide assets

Summer internship, RGL Associates, Sheffield
Summer 2018

- Developed professional and technical experience, early in my career, by undertaking finite element (FE) analysis of wind turbine bearings and delivering projects for a small engineering firm

Mechanical engineer, Project Sunbyte, University of Sheffield
January 2018 - Autumn 2018

- My student-lead team mounted a telescope system of our design on a helium balloon for capturing pictures of Solar flares to further the understanding of their risks, as part of the High-Altitude Student Platform of the NASA Balloon Program. The launch took place in New Mexico, USA.

Awards

- Milija Pavlovic Research Travel Fund award to attend the European Workshop on Structural Health Monitoring 2024
- Imperial College General fund and Old Centralians' Trust awards for attendance at the International Association on Bridge Maintenance and Safety 2024
- Awarded the 2nd Prize in the Imperial College PhD Summer Showcase 2023
- Invited to deliver an oral presentation at the 25th Young Researchers Conference 2023 of the Institution of Structural Engineers
- Awarded the Skempton PhD Scholarship

Skills

Languages: English and French (both native, C2), Italian and Spanish (A2)

Software: Microsoft, Rhino, Solidworks, PI System, Ansys FE modelling and computational fluid dynamics

Programming: Python, MATLAB, C, Linux command line (for High Performance Computing)

References

Dr Craig Buchanan, Senior Lecturer and my PhD supervisor, craig.buchanan@imperial.ac.uk

Professor Elizabeth Cross, Faculty Director of Research & Innovation at the University of Sheffield, e.j.cross@sheffield.ac.uk

Dr Thomas Reynolds, Senior Lecturer at the University of Edinburgh, t.reynolds@ed.ac.uk

List of publications

Glashier, T., Kromanis, R. and Buchanan, C. (2024) 'An iterative regression-based thermal response prediction methodology for instrumented civil infrastructure', *Advanced Engineering Informatics*, 60, p. 102347. doi: 10.1016/j.aei.2023.102347.

Glashier, T., Kromanis, R. and Buchanan, C. (2024) 'Temperature-based measurement interpretation of the MX3D Bridge', *Engineering Structures*, 305, p. 116736. doi: 10.1016/j.engstruct.2023.116736.

Glashier, T., Kromanis, R. and Buchanan, C. (2024) 'Temperature-based damage detection for the commissioning dataset of the MX3D Bridge', in *Proceedings of the 11th European Workshop on Structural Health Monitoring (EWSHM 2024), June 10-13, 2024*. Postdam, Germany: e-Journal of Nondestructive Testing. doi: 10.58286/29866.

Glashier, T., Buchanan, C. and Kromanis, R. (2024) 'Thermal response prediction of the MX3D bridge's operational dataset', in Jensen, J. S., Frangopol, D. M., and Schmidt, J. W. (eds) Bridge Maintenance, Safety, Management, Digitalization and Sustainability. London: CRC Press, pp. 2511–2519. doi: 10.1201/9781003483755-299.

Reynolds, T. P. S., **Glashier, T**., Cameron, J., Kromanis, R., Zhang, P., Wynne, Z., Tessier, A., Shuldiner, A., Cammers-Goodwin, S., Vahdatikhaki, F., Nagenborg, M., Siderius, K., Buchanan, C. (2024) 'Operational data collection and analysis for a smart 3D printed footbridge: the MX3D bridge', *Preprint submitted to Case Studies in Construction Materials.*

Glashier, T., Kromanis, R. and Buchanan, C. (2024) 'Preparation of long-term structural health monitoring data: Signal visualisation and processing', *Preprint submitted to Mechanical Systems and Signal Processing.*